

姓名：

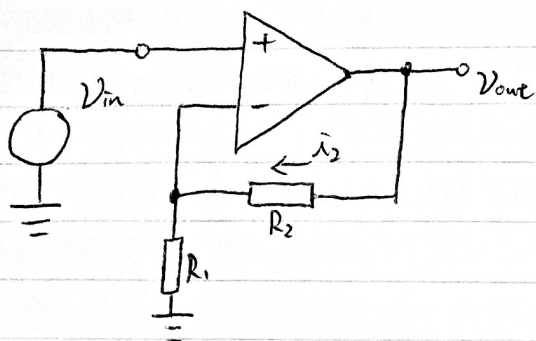
(號)

成績

年級：

科目：

作業一

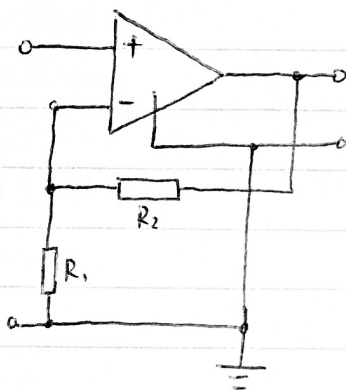


$$\frac{V_{out} - V_{in}}{R_2} = \frac{V_{in}}{R_1} = \frac{V_{out}}{R_1 + R_2}$$

$$V_{out} = V_{in} \left(1 + \frac{R_2}{R_1} \right)$$

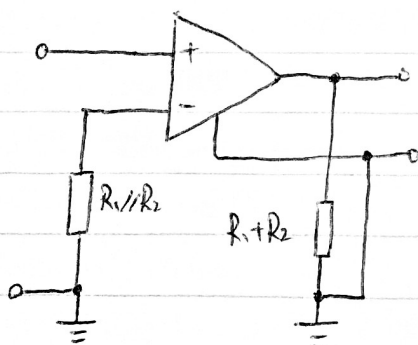
$$\Rightarrow A_v = 1 + \frac{R_2}{R_1}$$

負反饋連接方式：串並連接，電

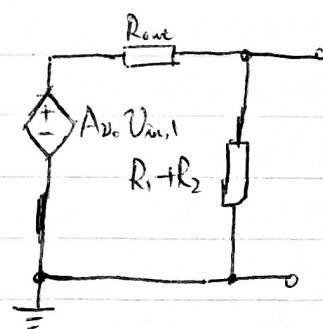
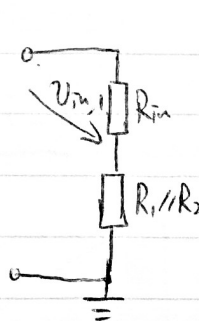


反饋係數：串並連接 h 相加， $h_{12} = \frac{V_1}{V_2} \Big|_{i_1=0} = \frac{R_1}{R_1 + R_2}$ (R_1 在 R_1 串 R_2 上的分壓比) $= \frac{1}{10} = 0.1$

開環放大器：



$\Rightarrow$



$$R_{in,0} = R_{in} + R_1 // R_2 = 2M\Omega + 0.9k\Omega = 2000.9k\Omega \Leftrightarrow G_{m,0} = 4.998 \times 10^{-4} S = 0.499mS$$

$$R_{out,0} = R_{out} // (R_1 + R_2) = 69.8\Omega$$

$$A_{v,0} = \frac{A_{vo}}{R_{out,0}} R_{in,0} = 5.33 \times 10^9 \quad A_{vo} = 20000$$

閉環放大器： $T = A_{vo} F = 20000$

$$G_{inf} = \frac{G_{m,0}}{1+T} = 25nS \Leftrightarrow R_{inf} = 40M\Omega, \quad R_{outf} = \frac{R_{out,0}}{1+T} = 3.49m\Omega$$

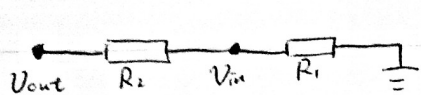
$$A_{vf} = \frac{A_{vo}}{1+A_{vo}F} \approx 9.99 \approx 10 = \frac{1}{F} \quad (\text{因為 } T = 20000 \gg 1)$$

女

作業2

如人體的血液流速調節。當人體運動後，血液會因為體熱變化而變化，穩定人的體溫

作業3

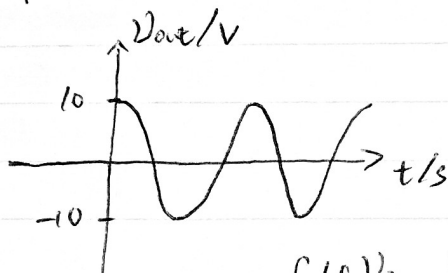


$$\frac{V_{out}}{R_1 + R_2} = \frac{V_{in}}{R_1} \Leftrightarrow V_{out} = V_{in} \left(1 + \frac{R_2}{R_1}\right) = 10 V_{in}$$

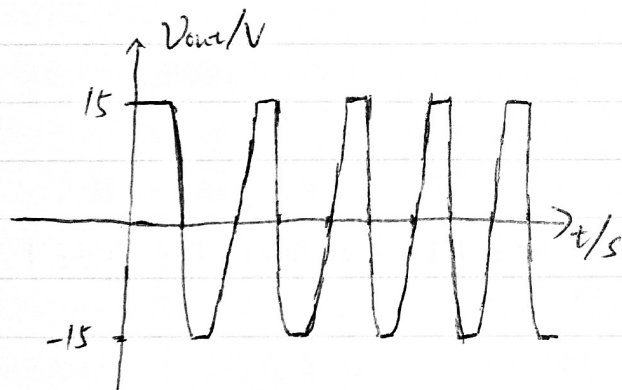
(當 $|V_{out}| \leq 15V$ 時, 即 $|V_{in}| \leq 1.5V$ 時)

當 $V_{in} = (1V) \cos(\omega t + \varphi)$ 時, $|V_{in}| \leq 1V < 1.5V$

$V_{ip} = 2V$

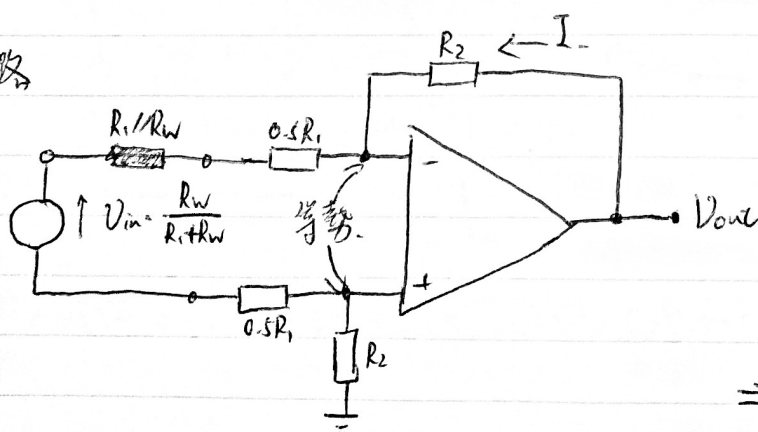


當 $V_{in} = (2.5V) \cos(\omega t + \varphi)$ 時, $V_{out} = \begin{cases} 10 V_{in}, & |\cos(\omega t + \varphi)| < 0.6 \\ \text{sgn}(\cos(\omega t + \varphi)) \cdot 15V, & |\cos(\omega t + \varphi)| > 0.6 \end{cases}$



作業四

等效電路



$$V_{out} \left(\frac{R_1}{2R_2} + \frac{R_1 R_w}{(R_1 + R_w) R_2} \right) = V_{in} \frac{R_w}{R_1 + R_w}$$

$$\Rightarrow V_{out} = \frac{1}{\frac{R_1(R_1 + R_w)}{2R_2 R_w} + \frac{R_1}{R_2}} V_{in}$$

$$V_{out} - IR_2 = IR_2 \Rightarrow I = \frac{V_{out}}{2R_2}$$

$$\begin{aligned} \cancel{I} V_{out} - V_{out} &= I(R_1 + 2R_2 + R_1 // R_w) - V_{in} \cdot \frac{R_w}{R_1 + R_w} \\ &= V_{out} \left(\frac{R_1}{2R_2} + \frac{R_1 // R_w}{R_2} + 1 \right) - V_{in} \cdot \frac{R_w}{R_1 + R_w} \end{aligned}$$

功能電壓放大

(當 $R_w \gg R_1, R_2$ 時)
 $A_v \approx \frac{2}{3} \frac{R_2}{R_1}$

培道中學校

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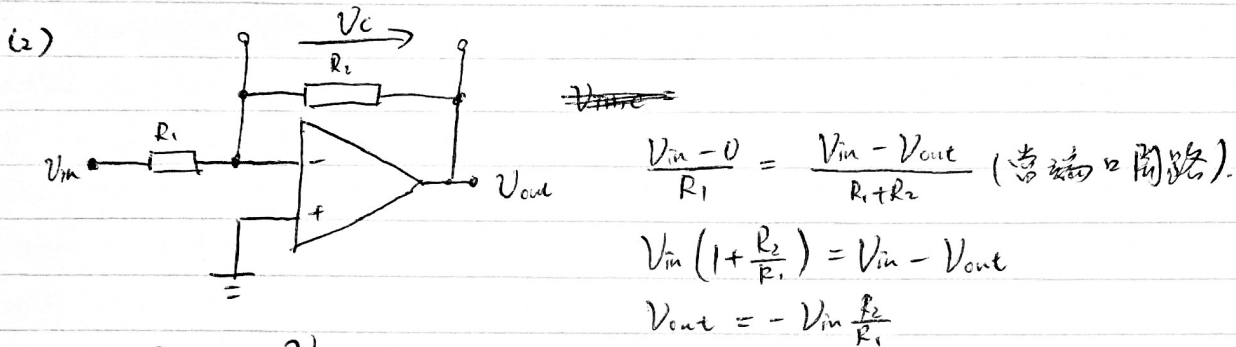
科目：

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作業 5

$$18. (1) \quad i_c = C \frac{dV_c}{dt} = -C \frac{dV_{out}}{dt} \Rightarrow \frac{V_{in}}{R_1} = -C \frac{dV_{out}}{dt} - \frac{V_{out}}{R_2}$$

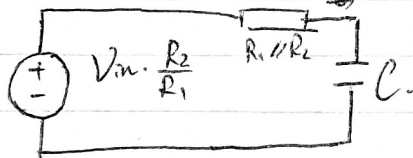
$$i_2 = -\frac{V_{out}}{R_2} \Rightarrow C \frac{dV_{out}}{dt} + \frac{V_{out}}{R_2} + \frac{V_{in}}{R_1} = 0.$$



$$V_{TH, C} = \frac{V_{in}}{R_1} \cdot R_2$$

$$R_{TH, C} = R_1 \parallel R_2 = \frac{R_1 R_2}{R_1 + R_2}$$

等效電路



$$V_c = i(R_1 \parallel R_2) + V_{in} \frac{R_2}{R_1}$$

$$= -C \frac{R_1 R_2}{R_1 + R_2} \frac{dV_c}{dt} + V_{in} \frac{R_2}{R_1}$$

$$C \frac{R_1 R_2}{R_1 + R_2} \frac{dV_c}{dt} + V_c = V_{in} \frac{R_2}{R_1}$$